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ORIGINAL ARTICLE

Complications after reoperative thyroid surgery: retrospective evaluation of 152 consecutive cases

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Abstract

Reoperative thyroid surgery is an uncommon procedure that is indicated in recurrent benign or malignant disease. It is associated with a high complication rate, especially of hypoparathyroidism and recurrent nerve palsy. We retrospectively reviewed our series of patients on whom reoperative thyroid surgery was performed and we compared this group with patients who underwent primary thyroidectomies. From 2002 to 2015, 4572 thyroidectomies were performed at our institution; among these, 152 (3.3%) were for benign or malignant recurrent disease. We observed a higher rate of transient hypoparathyroidism in secondary vs primary surgery (56.6% vs 25.9%; $p < 0.0001$), of permanent hypoparathyroidism (10% vs 2.0%; $p < 0.0001$) and of transient recurrent nerve injury (4.6% vs 1.4%; $p < 0.05$). Reoperative thyroid surgery is a technical challenge with a high incidence of complications. Scarring, edema, and friability of the tissues together with distortion of the landmarks make reoperative surgery hazardous. Careful assessment of patient's risk factors, physical examination, and if necessary fine needle aspiration cytology are crucial for selecting the patients who should undergo reoperation. Research registry n. 2617 registered 5 June 2017 (retrospectively registered).

Keywords Reoperative thyroid surgery · Total thyroidectomy · Hypoparathyroidism · Recurrent nerve injury

Background

The treatment for nodular benign thyroid diseases has varied over the years and it is still a matter of debate. Subtotal thyroidectomy has been the treatment of choice for many years, because it was thought to be a safer procedure despite the risk of recurrent nodular goiter, which ranges from 3 to 43% [1–6]. Suppressive L-thyroxine therapy has been used to reduce the rate of recurrence but with controversial efficiency [6–11].

Reoperative thyroid surgery is performed in case of recurrent single nodule or multinodular goiter in case of compressive symptoms, suspected malignancy or in some cases of recurrent thyrotoxicosis [10, 12].

Reoperative thyroid surgery is associated with higher complication rates because of scar tissue, which makes difficult to recognize and preserve neck structures such as recurrent laryngeal nerves or the vascular pedicles of the parathyroid gland [13].

We retrospectively reviewed our series of patients on whom reoperative thyroid surgery was performed.

Materials and methods

After ethical approval by local ethics committee, we performed a retrospective study on patients who had undergone thyroidectomy at our Department of General and Endocrine Surgery (University of Cagliari) between September 2002 and December 2015. After receiving Institutional Review

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Board approval, patients' demographics, preoperative data, pathological findings, and postoperative complications (including recurrent nerve injury and hypoparathyroidism) were recorded. Patients were divided into two groups depending on if they had undergone primary or secondary thyroidectomy.

Patients were included in the second group if they had previously undergone a lobectomy or an enucleation of a thyroid nodule, and if they had then a secondary procedure consisting in a totalization (in case of previous enucleation) or in a resection of a recurrent goiter or cancer. Indications for reoperative thyroid surgery were recurrent multinodular goiter with compressive symptoms, recurrent thyrotoxicosis, or suspected malignancy assessed by cytology in patients with the previous partial resection of a lobe. Patients with the previous resection of a lobe and secondary contralateral lobectomy were included in the first group.

Preoperative indirect laryngoscopy was performed on all patients.

Secondary thyroidectomy was performed using a standard operative technique. Depending on the amount and the density of fibrous tissue, a lateral or medial approach was performed to access the thyroid lodge. In the presence of dense scar tissue, dissection was begun laterally by mobilizing sternocleidomastoid muscles and dividing infrahyoid muscles to avoid fibrous tissue around the thyroid remnant when the lobe had been partially removed during the previous surgery. Otherwise, a medial approach was performed. The first step was ligation of the superior thyroid vessels with special attention being paid to the superior parathyroid gland and the external branch of the superior laryngeal nerve. Then, the recurrent laryngeal nerve was identified in the lower part of the neck and traced very carefully along its cervical course. Intrathoracic goiters extending into the superior mediastinum were removed by progressive traction through the collar incision. A median sternotomy was necessary in one patient with a large recurrent multinodular goiter with substernal extension.

The inferior parathyroid gland was preserved with its vessels when it was located far from thyroid lobe. Any parathyroid gland that could not be salvaged because its anatomic location was minced and placed in a sternocleidomastoid muscle.

Since 2012, we routinely used intermittent intraoperative neuromonitoring: in these cases, the patients were intubated with a Nerve Integrity Monitor (NIM) Standard Reinforced Electromyography (EMG) Endotracheal Tube (Medtronic Xomed, Jacksonville, FL, USA) and a Prass monopolar stimulation probe (Medtronic Xomed) was used for nerve stimulation during surgery. At the beginning of surgery, stimulation was applied at a level corresponding to the vagus nerve to ensure that the monitoring system was working; later, stimulation was applied to the structures believed to be the RLN. After thyroid removal and complete hemostasis, stimulation was applied to vagus nerve and to RLN to assess integrity of RLN.

If the course of reoperation was uneventful, the patient was discharged from the hospital on postoperative day 2. Indirect laryngoscopy was repeated in patients with postoperative dyspnoea, hoarseness, or loss of voice quality. Serum calcium and phosphorus levels were measured 3 h after surgery and daily, while patients were hospitalized, and then weekly postoperatively until normalized. No prophylactic calcium was administrated. If low serum calcium was found any time after surgery (< 8.0 mg/dl), PTH measurement was performed. Hypoparathyroidism was defined as serum PTH lower than 14 pg/ml. If hypocalcemia was found, oral calcium was administrated. Hypoparathyroidism was defined permanent if lasted more than 6 months after surgery.

All patients were submitted to a minimum follow-up of 6 months.

Chi-squared test was used for categorical data. Results were considered statistically significant if p value was ≤ 0.05. Continuous data are reported as the mean value $\pm$ standard error of the mean. Calculations were performed with MedCalc® 12.7.0.0.

Results

From September 2002 to December 2015, we performed in our institution 4572 thyroid operations. 207 (4.5%) patients had undergone previous thyroid surgery: 55 had undergone secondary contralateral lobectomy and were included in the first group, while 152 had a thyroid reoperation in a previously dissected thyroid lodge and were included in the second group.

Within the second group, 24 (15.8%) had undergone preoperative fine needle aspiration cytology (FNAC) and had a cytologic diagnosis of suspected malignancy or malignancy. In the other cases, indication for reoperative surgery was recurrent multinodular goiter in 122 patients (80.3%) and recurrent thyrotoxicosis in 6 (3.9%). In the second group, primary surgery consisted in lobectomy in 118 cases (77.6%) and in enucleation of a thyroid nodule in 34 (22.4%).

There were 136 (89.4%) women and 16 (10.6%) men with a mean age of 53.4 ± 14.4 years. Five (3.3%) of the initial operations in these patients were performed by our team and 147 (96.7%) were by surgeons at other centres. All but two patients had undergone a single prior operation, two patients' two prior operations. The median interval between the original operation and the reoperation was 194 months (range 1 month–45 years).

Preoperative indirect laryngoscopy revealed a RLN injury due to previous surgery in 2 (1.3%) patients.

Bilateral completion thyroidectomy was performed in 90 (59.2%) cases (in 5 associated with lymphadenectomy and in 3 to parathyroidectomy), lobectomy in 58 (38.2%) cases, removal of a mediastinal recurrence in 2 (1.3%) and removal of a pyramidal remnant in 2 (1.3%).

Histological examination revealed papillary cancer in 23 (15.1%) patients (7 microcarcinomas) and follicular cancer in 3 (2%). In 16 (10.5%) cases, malignant neoplasm was preoperatively unsuspected, and in 10 (6.5%) cases, preoperative suspect of papillary cancer was confirmed. In the remaining 126 (82.9%) patients, histological examination demonstrated a benign disease.

Mean operative time was 130.7 min (range 55–270).

Full comparison between primary and secondary thyroidectomy is reported in Table 1. Indications for primary thyroidectomy were more likely to be for hyperthyroidism and for thyroid carcinoma than for secondary thyroidectomies. Reoperative surgery required a longer operative time. There were no differences regarding postoperative stay.

Temporary hypoparathyroidism occurred in 51 (56.6%) patients and definitive hypoparathyroidism in 9 (10%) out of 90 patients who underwent bilateral secondary surgery. In patients submitted to primary thyroidectomy incidence was 25.9% ($p < 0.0001$) and 2.0% ($p < 0.0001$), respectively (Table 2).

Transient unilateral recurrent laryngeal nerve palsy occurred in 7 (4.6%) cases and permanent unilateral nerve palsy in 1 (0.7%). In primary thyroidectomies, incidence was 1.4% ($p = 0.013$) and 0.2% ($p = 0.82$), respectively (Table 1). We observed bilateral nerve palsy in 4 (0.1%) cases of primary thyroidectomies and no one in secondary surgery.

NIM was used in 41 (26.9%) cases of reoperative thyroidectomies; in these patients, we observed 2 (4.8%) nerve injuries. In the remaining 111 (73.1%) cases, where NIM was not used, we found 6 (5.4%) nerve injuries ($p = 0.78$; Table 3).

In 4 (2.6%) cases, surgical revision was necessary for postoperative haemorrhage. In all cases, bleeding happened

Table 1 Comparison between primary and secondary thyroidectomies

	Primary thyroidectomies ($n = 4420$)	Secondary thyroidectomies ($n = 152$)	p value
Sex (male:female)	511:3909	16:136	0.79
Age (years)	55.3 ± 21.2	53.4 ± 14.4	0.67
Hyperthyroidism	613 (13.8%)	6 (3.9%)	$p < 0.01$
Autoimmune thyroiditis	1954 (44.2%)	62 (40.8%)	0.45
Monolateral surgery	115 (2.6%)	62 (40.8%)	$p < 0.01$
Bilateral surgery	4305 (97.4%)	90 (59.2%)	
Central compartment lymph node dissection	411 (9.3%)	5 (3.3%)	$p < 0.01$
Lateral neck lymph node dissection	41 (0.9%)	0	0.45
Operative time (min)	97.5 ± 28.7	130.7 ± 41.3	$p < 0.01$
Histolopathological diagnosis			
Bening thyroid disease	3125 (70.7%)	126 (82.9%)	$p < 0.01$
Differentiated thyroid carcinoma	1295 (29.3%)	26 (17.1%)	
Weight of thyroid	41.6 ± 42.7	31.7 ± 38.5	0.16
Postoperative stay (days)	2.8 ± 1.9	2.4 ± 2.1	0.64

Table 2 Complications after reoperative thyroid surgery compared with primary thyroidectomies

	Primary thyroidectomies ($n = 4420$)	Secondary thyroidectomies ($n = 152$)	p value
Unilateral recurrent laryngeal nerve palsy			
Transient	1.4% (61)	4.6% (7)	0.013
Permanent	0.2% (10)	0.7% (1)	0.82
Bilateral recurrent laryngeal nerve palsy	4 (0.1%)	0	–
Hypoparathyroidism			
Transient	25.9% (1148)	56.6% (51)[a]	< 0.0001
Permanent	2.0% (88)	10% (9)[a]	< 0.0001
Postoperative haemorrhage	1.4% (63)	2.6% (4)	0.38

[a]Over 90 patients who underwent bilateral surgery

Table 3 Use of NIM system in secondary thyroidectomies

	Use of NIM (41)	Not use of NIM (111)	*p* value
Recurrent laryngeal nerve palsy	2 (4.8%)	6 (5.4%)	0.78

NIM nerve intraoperative monitoring

in the first 6 h and we had intraoperative findings of diffuse haemorrhage without evidence of a major vessel bleeding. In primary thyroidectomies, incidence of postoperative haemorrhage requiring surgical revision was 1.4% ($p=0.38$) (Table 1).

Discussion and commentary

Reoperative thyroid surgery is an uncommon and challenging procedure. In our series, most of the reoperations were for patients that had been operated on in different centres; this could be explained by the fact that ours is a high-volume endocrine centre, thus tough cases are referred to our department from all over the region.

The incidence of complications during secondary thyroid surgery is high [13–16]. Scarring, edema, and friability of the tissues, together with distortion of the landmarks, make reoperative surgery hazardous [17, 18]. The complications are mainly related to the extent of the recurrent disease, the proximity of disease to the recurrent laryngeal nerve and the indications for which reoperative surgery is performed, along with scarring and fibrosis [13].

In our series, the main indication for reoperative surgery was for benign disease; this is probably related to the fact that ours is an iodine-deficiency region with high rates of multinodular goiter and autoimmune thyroiditis.

The risk of complications varies according to the initial surgical resection: a higher risk of complications is described when previous surgery has been performed on both sides [12, 19]. In addition, the risk of complications has been described to increase with the number of reoperations [10]. In our experience, bilateral surgery is associated with a higher risk of complications, particularly for postoperative bleedings and hypoparathyroidism, even though differences were not statistically significant.

Recurrent laryngeal nerve injury is the most feared complication [20]. The risk of a recurrent laryngeal nerve palsy has been shown to be up to eight times higher after operation for relapses than after the primary operation [2, 21]. The incidence of temporary and permanent palsy of recurrent laryngeal nerve may be as high as 20% in recurrent goiters [2, 3, 17]. Reoperative thyroid surgery had more frequent recurrent laryngeal nerve injuries, both temporary (0–22%) and permanent (0–13%) [10, 22]. Hyperthyroidism and not exposing the nerve during operation seems to be predisposing factors [12, 21]. In our experience, transient recurrent laryngeal nerve palsy was nearly four times higher than for primary operations and differences were highly statistically significant. Permanent palsy was also higher for secondary surgery, but the difference was not statistically significant.

A limit of our study is that indirect laryngoscopy was not performed on all patients postoperatively, but only on selected symptomatic patients: therefore, the incidence of temporary RLN injury is probably underestimated as some patients compensate quite well, presenting minimal or very vague symptoms in the immediate postoperative period.

Another feared complication of repeated thyroid surgery is permanent hypoparathyroidism that can be a very disabling complication [20]. It has been reported that the incidence of transient hypoparathyroidism was 0–25% and of definitive hypoparathyroidism from 0 to 22% for reoperative thyroid surgery [9, 10, 20, 22–24]. In some cases, the parathyroid glands may be hidden by scar tissue from the previous operative procedure [20]. In our experience, the incidence of transient and definitive hypoparathyroidism was very high (56.6% and 10%) compared with primary thyroidectomies (25.9% and 2%) and the difference was statistically significant.

Some authors reported that mortality after repeat operation is higher; in one case reaching 3% [16, 21], we had no mortality in our experience.

In repeat surgery for recurrences, the lateral approach is popular to avoid the infiltration by connective tissue of Kocher's approach [10, 12, 25] and because it allows identification of the nerves in a previously undissected area [26]. Rather than reopening the median raphe, the straps are retracted medially and a plane anterior to the sternocleidomastoid is entered [10]. The recurrent laryngeal nerve should be always identified, usually in the lower neck, then followed along its cervical course [10]. However, no significant difference in the rate of palsy between the lateral and medial approach was reported. A tendency for a slightly higher rate of palsy using the lateral approach was reported in one study [10]. A possible explanation is that the lateral approach avoids the connective tissue ventrally, but dorsally, it cannot be avoided by either approach, where the nerve is at risk; moreover, surgeons usually have less experience using the lateral approach, and hence, the surgical field and anatomy may be less familiar [10].

Intraoperative neuromonitoring is useful in reoperative thyroid surgery especially in situations in which the anatomic situation diverges from the normal. Intraoperative neuromonitoring may reduce the morbidity of reoperative thyroid surgery [1, 17]. We routinely use intraoperative neuromonitoring during thyroid surgery since 2012 and we recently reported a large series with nearly 1500 patients

[27]. In our opinion, intraoperative neuromonitoring may be particularly useful for reoperative thyroid surgery, making the detection of the nerve in the scar tissue easier, although we still do not have enough data to verify if this approach could significantly reduce nerve injury rate.

Other described uncommon complications are chylotorax and severe bleeding from major vessels (common carotid artery) [16].

Total thyroidectomy is the most effective operation for preventing recurrence of benign multinodular goiters [10, 28, 29]. In addition to disease recurrence, numerous studies have demonstrated that a significant proportion of patients who have surgery for multinodular goiter harbour occult malignancy. Removing all thyroid tissue essentially eliminates the risk of developing recurrent disease and thyroid carcinoma in the remnant tissue [4]. In our experience, histological examination revealed malignancy in 23 patients (15.1%), in 16 cases (10.5%), malignant neoplasm was unsuspected, and in 10 (6.5%) cases, preoperative suspect of papillary cancer was confirmed.

To reduce the need of reoperative surgery, a total thyroidectomy should also be considered in patients with suspicious nodules and multiple lesions, hyperplastic nodular goiter or thyroiditis, because in these patients, the rate of malignancy is higher [30, 31].

Conclusions

Reoperative thyroid surgery is a challenge, and has a high rate of complications because of scarring and loss of normal tissue planes. For this reason, the initial treatment should be carefully evaluated and every effort should be made to avoid recurrent disease. Careful assessment of patient's risk factors, physical examination, and if necessary, fine needle aspiration cytology is crucial for selecting the patients who should undergo reoperation.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Research involving human participants and/or animals All data was anonymized and de-identified prior to analysis and the study was reported in line with the declaration of Helsinki.

Ethical standards Ethical approval was given by local ethics committee (Independent Ethics Committee, University of Cagliari).

Informed consent Each patient included in the study gave consent to participate to this study.

Availability of data and materials The data sets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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